R.M.K. COLLEGE OF ENGINEERING AND TECHNOLOGY
(AN AUTONOMOUS INSTITUTION)
R.S.M. NAGAR, PUDUVOYAL - 601206

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

PROJECT REPORT

on

NALAMMESH – OFFLINE-FIRST RURAL HEALTHCARE PLATFORM

Submitted by

THARUN S	111624102222
YOGESH S	111624102246
VEERAMANIKANDAN	111624102236
VISHNU VARDHAN S	111624102240

COURSE: PRODUCT DEVELOPMENT LAB - III

COURSE CODE: 24AI511

OCTOBER 2026

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Certified that this project report “NALAMMESH – OFFLINE-FIRST RURAL HEALTHCARE PLATFORM” is the bonafide work of “THARUN S” (111624102222), “YOGESH S” (111624102246), “VEERAMANIKANDAN” (111624102236) and “VISHNU VARDHAN S” (111624102240), who carried out the project work under my supervision.

SIGNATURE

Dr. R. RAJA GURU

HEAD OF THE DEPARTMENT

SIGNATURE

SUPERVISOR

Submitted for the Viva Voce examination held on _____.

INTERNAL EXAMINER

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Dr.R.RAJA GURU M.Tech., Ph.D
Professor & Head
Department of Computer Science & Engineering
R.M.K. College of Engineering & Technology
R.S.M. Nagar, Puduvoyal - 601 206

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Department of Computer Science & Engineering
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R.S.M. Nagar, Puduvoyal - 601 206

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PO9: Communication: Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences

PO10: Project Management and Finance: Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.

PO11: Life-Long Learning: Recognize the need for, and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change. (WK8)

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02/09/2026

HoD

Dr.R.RAJA GURU M.Tech., Ph.D
Professor & Head
Department of Computer Science & Engineering
R.M.K. College of Engineering & Technology
R.S.M. Nagar, Puduvoyal - 601 206

ABSTRACT

Rural public healthcare in India depends on a chain of facilities: the Sub-Centre in the village, the Primary Health Centre (PHC), the Community Health Centre (CHC) and the District Hospital. Patients who need specialist care must move up this chain, yet in most districts their details are still written in paper registers, referrals travel as slips of paper with the patient, and the receiving hospital learns about a critical case only when the ambulance reaches the gate. Mobile networks in these areas are weak or absent for long periods, so ordinary online systems stop working exactly where they are most needed. As a result, patient history is lost between facilities, emergencies wait without anyone being alerted, and district officers have no reliable view of how referrals are handled.

NalamMesh is an offline-first digital public healthcare platform built to solve these problems. Health workers such as the Auxiliary Nurse Midwife (ANM), the Medical Officer (MO) and the District Health Officer (DHO) sign in with a staff ID and PIN and see only the screens their role allows. A health worker registers a patient, records vital signs and danger signs, and receives a RED, YELLOW or GREEN triage result from an on-device TensorFlow.js model backed by a rule engine based on the Indian Public Health Standards (IPHS); the danger-sign rules can only raise urgency, never lower it, and the clinician always decides. A referral can then be sent to a higher facility. Every record is saved first in the device's own IndexedDB database, so the application keeps working without a network, and uploads wait in a durable outbox until a connection returns.

When a connection is available, a Socket.io mesh relay carries live referrals between facilities and a FastAPI district record service shows the receiving facility a Pre-Arrival Board before the patient arrives. Every request for identified patient records is written to an access log that the DHO can review. The system was tested with 38 automated verification suites and browser checks covering sign-in, role permissions, triage cases, the referral flow, offline start-up, accessibility and a complete production flow. NalamMesh reduces delay in emergency referrals, keeps a complete and audited patient record across facilities, and helps the health department deliver continuous and accountable care to rural communities.

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LIST OF ABBREVIATIONS

1. ANM: Auxiliary Nurse Midwife
2. MO: Medical Officer
3. DHO: District Health Officer
4. PHC: Primary Health Centre
5. CHC: Community Health Centre
6. DH: District Hospital
7. IPHS: Indian Public Health Standards
8. ABDM: Ayushman Bharat Digital Mission
9. ABHA: Ayushman Bharat Health Account
10. FHIR: Fast Healthcare Interoperability Resources
11. OPD: Out-Patient Department
12. PWA: Progressive Web Application
13. API: Application Programming Interface
14. TF.js: TensorFlow.js
15. PIN: Personal Identification Number
16. WCAG: Web Content Accessibility Guidelines
17. GIGW: Guidelines for Indian Government Websites
18. DPDP: Digital Personal Data Protection
19. HFR: Health Facility Registry
20. HPR: Healthcare Professionals Registry

CHAPTER 1

INTRODUCTION

Access to timely healthcare remains one of the biggest challenges for rural India. A large share of the population lives in villages that are far from hospitals, and the public health system reaches them through a tiered network of facilities. When this network works well, a patient seen by an ANM at the Sub-Centre can be referred to the PHC, then to the CHC and finally to the District Hospital, with each level adding doctors, equipment and beds. In practice, the movement of patients and information across these levels is slow and unreliable. This project, NalamMesh, applies modern web technology to make this continuum of care faster, safer and accountable, while working within the realities of poor connectivity in rural areas.

1.1 BACKGROUND

India's rural public health system is organised under the National Health Mission. A Sub-Centre, now often run as an Ayushman Arogya Mandir, serves a few villages and is staffed by an ANM and community health workers. A PHC serves a larger population with a Medical Officer, a small laboratory and a few beds. The CHC acts as the first referral unit with specialists, an operation theatre and more beds, and the District Hospital provides advanced care. Each facility keeps registers for out-patients, pregnant women, immunisation, stock and referrals, and most of these registers are still maintained on paper.

Several national programmes have introduced digital tools, such as the Ayushman Bharat Digital Mission (ABDM) with its ABHA health ID, the eSanjeevani teleconsultation service and various reporting portals. However, these tools generally assume a stable internet connection. In many blocks the mobile signal drops for hours, power supply is irregular, and health workers share a small number of devices. Under these conditions an online-only application cannot be relied on at the moment of an emergency, and staff fall back to paper and phone calls.

1.2 PROBLEM STATEMENT

When a pregnant woman with danger signs, a child with severe malnutrition or a snakebite victim arrives at a Sub-Centre, the ANM must decide quickly how urgent the case is and where to send the patient. Today the decision depends entirely on the individual worker's experience, the referral is written on a slip, and the receiving facility is informed, if at all, by a phone call. The main problems with this process are listed below.

- Patient history is lost between facilities because paper records stay where they were written.
- There is no proof that a referral was sent, received or acted upon, and nobody is alerted when an emergency is ignored.
- The receiving facility cannot prepare a bed, blood or a specialist before the patient arrives.
- Triage decisions vary between workers and danger signs can be missed under pressure.
- District officers have no live view of pending emergencies, bed occupancy or referral delays.
- Online systems fail in areas with weak networks, so digital records are incomplete.

1.3 PURPOSE OF NALAMMESH

The purpose of NalamMesh is to make sure that every patient's record and referral travels with them across the continuum of care, even when the network is unavailable. The name combines 'Nalam', a Tamil word meaning well-being, with 'mesh', the network of facilities that the platform connects. The application runs on the health worker's phone or computer, stores everything on the device first, and synchronises automatically whenever a connection becomes available.

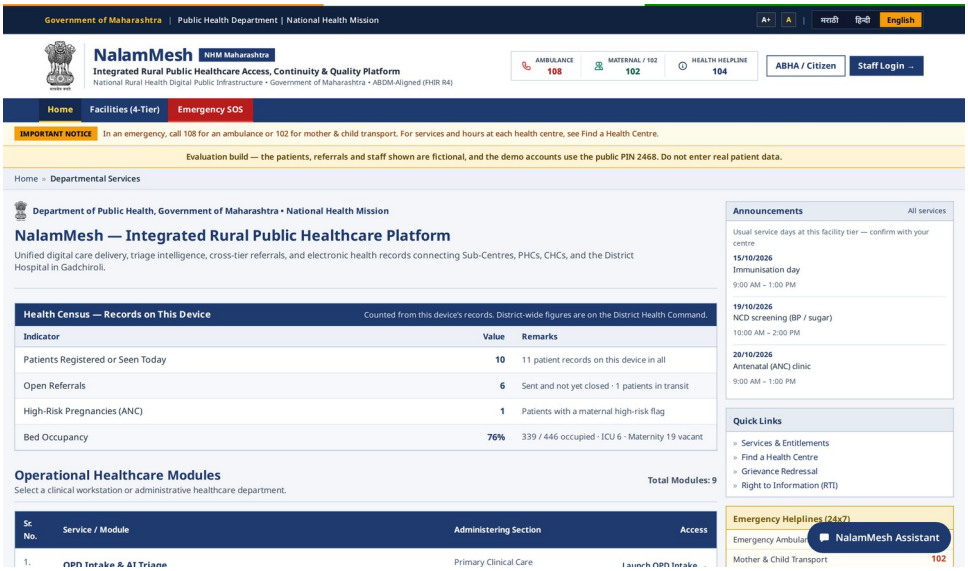


Figure 1.1: NalamMesh home page

1.3.1 IMPROVING EMERGENCY REFERRALS

In an emergency, minutes matter. NalamMesh lets the referring worker send the patient's triage result, vitals and clinical summary to the receiving facility as soon as the referral is created. The receiving Medical Officer sees the patient on the Pre-Arrival Board, can acknowledge the referral and hold a bed, and the referring worker sees each status change. If an emergency referral is not acknowledged within ten minutes, it is escalated to the District Health Officer so that it cannot be silently ignored.

1.3.2 SUPPORTING HEALTH WORKERS

Front-line health workers handle a wide range of cases with limited support. NalamMesh gives them a structured registration form, a decision-support triage result that highlights danger signs, follow-up lists for pregnant women and chronic patients, and an in-app assistant that answers questions about the platform and rural public health from data stored on the device. The triage slip clearly states that it is decision support and that the health worker or Medical Officer makes the final decision.

1.4 OBJECTIVES

The main objective of this project is to design and develop a reliable, secure and offline-capable system for patient registration, triage and referral across rural health facilities. The specific objectives are:

- To provide a single application that works fully offline and synchronises automatically when a network is available.
- To allow secure sign-in with a staff ID and PIN, with role-based access for six roles.
- To capture complete patient details such as demographics, ABHA ID, vital signs, danger signs and chief complaint.
- To give a RED, YELLOW or GREEN triage result using an on-device AI model and IPHS clinical rules that can only raise urgency.
- To send referrals between facilities and track their status until the patient is discharged.
- To show the receiving facility a Pre-Arrival Board so that it can prepare before the patient arrives.
- To record every access to identified patient data in an access log reviewed by the District Health Officer.

1.5 SCOPE OF THE PROJECT

The project covers the complete flow from patient registration at a Sub-Centre or PHC to admission and discharge at the receiving facility. It includes the device application for browsers and Android, the mesh relay server, the district record service and its database, and the automated tests that verify them. Ambulance dispatch, audio and video teleconsultation and payment are outside the scope; ambulances are still called through 108 and 102, and teleconsultation uses eSanjeevani alongside a structured record in the app.

CHAPTER 2

LITERATURE REVIEW

This chapter reviews existing work on digital health records, referral systems, triage tools and offline-first software, and identifies the gaps that NalamMesh addresses. The review covers government platforms, research on clinical decision support in low-resource settings, and the technologies used to build applications that work without a reliable network.

2.1 DIGITAL HEALTH RECORDS IN INDIA

The Ayushman Bharat Digital Mission aims to build a national digital health ecosystem with a unique ABHA health ID for every citizen, registries of facilities (HFR) and professionals (HPR), and a framework for exchanging records in the FHIR R4 standard. State governments and programmes such as RCH and NCD screening maintain their own portals. These systems mainly record data after care has been given and are designed for reporting; they are rarely used at the point of care in a village, where the network is weakest.

2.2 REFERRAL MANAGEMENT SYSTEMS

Studies of referral systems in low- and middle-income countries consistently report missing referral slips, lack of feedback from the higher facility, and patients arriving without the receiving team being informed. Mobile referral applications developed in pilot projects improved communication, but most of them depended on SMS or a continuous data connection and did not maintain the full patient record across facilities. Few provided a mechanism to escalate an emergency that had not been acknowledged.

2.3 CLINICAL TRIAGE AND DECISION SUPPORT

Structured triage protocols such as the WHO Emergency Triage Assessment and Treatment (ETAT) guidelines and the danger signs listed in IPHS and maternal health protocols help health workers identify critical patients. Machine-learning

models have been proposed to predict urgency from vital signs, but researchers warn that such models must not override clear danger signs and must be validated clinically before use. A safe design therefore combines a model with deterministic rules and keeps the clinician responsible for the decision.

2.4 OFFLINE-FIRST AND MESH TECHNOLOGIES

Offline-first applications store data locally and synchronise later. Modern browsers provide IndexedDB for structured storage and service workers that can cache the application so that it opens without a network. Progressive web applications can be installed like native apps, and wrappers such as Capacitor package them for Android. Real-time communication libraries such as Socket.io allow devices to exchange events through a relay when a connection is available, and fall back gracefully when it is not.

2.5 COMPARISON OF EXISTING APPROACHES

Table 2.1: Comparison of existing approaches with NalamMesh

Approach	Works offline	Referral tracking	Decision support	Audit of access
Paper registers and slips	Yes	No	No	No
Online reporting portals	No	Partial	No	Partial
SMS-based referral apps	Partial	Yes	No	No
Stand-alone triage apps	Yes	No	Yes	No
NalamMesh	Yes	Yes, with escalation	Yes, rules + model	Yes

2.6 CHALLENGES IDENTIFIED

The review shows that existing solutions solve parts of the problem but not the whole. Online systems fail in low-connectivity areas; offline tools do not share records across facilities; triage tools are not linked to referrals; and very few systems restrict who can read identified patient records or keep a log of each access, which is required under the Digital Personal Data Protection Act, 2023.

2.7 CONCLUSION

The literature review concludes that there is a need for a single platform that works offline at the point of care, links triage to referrals, informs the receiving facility before the patient arrives, escalates ignored emergencies, and protects and audits patient data. NalamMesh was designed to meet these requirements.

CHAPTER 3

SYSTEM DESIGN AND FUNCTIONALITY

This chapter describes the architecture of NalamMesh, the technologies used, and the main functions of the system from the point of view of each type of user.

3.1 SYSTEM ARCHITECTURE

NalamMesh follows an offline-first, device-centred architecture. The health worker uses the app on a phone or computer, where every record is first saved in IndexedDB. The app talks to the mesh relay over WebSocket for sign-in and live referrals, and uploads records to the district record service over HTTPS when a connection is available. The receiving facility reads incoming patients from the district service on its Pre-Arrival Board, and the district service stores records in its database and logs every access. If the relay or the district service is unreachable, care continues on the device and the uploads wait in the outbox.

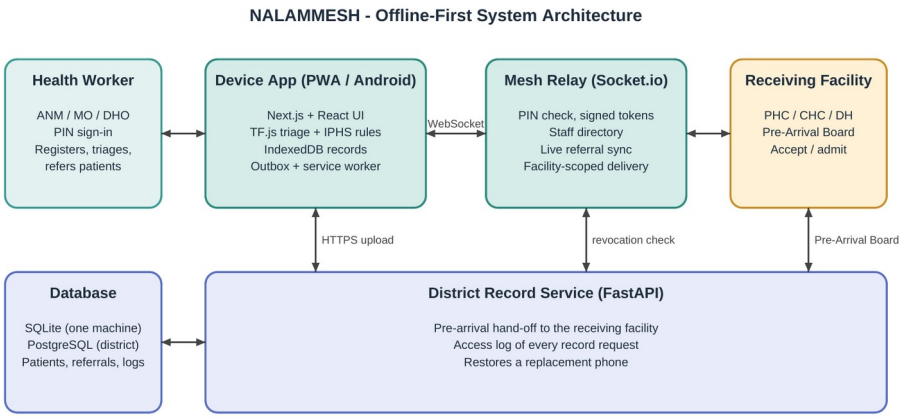


Figure 3.1: Architecture of the NalamMesh platform

3.2 TOOLS AND TECHNOLOGIES

Table 3.1: Tools and technologies used

Component	Technology	Purpose
Front end	Next.js 15, React 19, TypeScript, Tailwind CSS	User interface for sign-in, dashboards, triage, referrals and boards
State and storage	Zustand, IndexedDB	App state and offline storage of patients, queue, referrals and stock
AI triage	TensorFlow.js, IPHS rule	On-device RED / YELLOW / GREEN triage

	engine	
Mesh relay	Node.js, Socket.io	PIN sign-in, signed tokens, live referral sync
Back end	Python, FastAPI, SQLAlchemy	District record service, access log, record restore
Database	SQLite, PostgreSQL 16	Uploaded records, referrals and access log
Offline and mobile	Service worker, Capacitor	Installable app that opens offline; Android build
Testing	Playwright, axe-core, verify suites	End-to-end, offline, accessibility and unit checks

3.3 USER ROLES

NalamMesh defines six roles. Every screen, action and server route checks both the role and the facility to which the staff member is posted, so that, for example, a PHC Medical Officer reads only the pre-arrival board of their own PHC.

Table 3.2: User roles and their main functions

Role	**Main functions**
ANM / Nurse	Registers patients, records vitals, triages, refers, follows up pregnant women and chronic patients
Medical Officer	OPD triage, queue, referrals, Pre-Arrival Board, diagnostics and medicine stock
Hospital Admin / Bed Manager	Beds and equipment, admissions and discharges at the receiving facility
Specialist Doctor	Reviews referred patients and teleconsultation records
District Health Officer	District Command Centre, escalations, Data Inspector and Audit Trail
Super Admin	Creates and deactivates staff accounts and assigns postings

3.4 PATIENT REGISTRATION AND TRIAGE

The OPD Patient Registration and Digital Triage screen collects the patient's name, age, gender, ABHA ID, village, pregnancy or child status, and the vital signs measured by the health worker: oxygen saturation, pulse, blood pressure, temperature and respiratory rate, together with danger signs and the chief complaint. When the worker runs triage, the on-device model estimates urgency and the IPHS rule engine checks every danger sign. The rules can only raise the urgency; with the

model unavailable, the rules alone decide. The result is shown as RED, YELLOW or GREEN with the reasons and a slip that states the clinician decides.

3.5 REFERRAL WORKFLOW

A referral records the patient, the referring and receiving facilities, the urgency and a clinical summary. It moves through a defined life cycle, and each change is visible to both facilities and to the DHO.

Table 3.3: Referral life cycle

Step	Status	Description
1	Sent	Health worker refers a triaged patient; the record is queued for upload
2	Acknowledged	Receiving facility has seen the referral; unacknowledged emergencies escalate to the DHO
3	Accepted	Bed is held and the patient appears on the Pre-Arrival Board while in transit
4	Arrived / Admitted	Patient reaches the facility and is admitted
5	Discharged	Treatment is complete and the referral is closed

3.6 OFFLINE STORAGE AND SYNCHRONISATION

Every change made on the device is written to IndexedDB and added to a durable outbox. When a connection is available, the outbox uploads records oldest first and never drops an item; failed uploads are retried. The service worker pre-caches every screen and the build's own files, so an installed app opens with the server stopped. A replacement phone can restore its facility's records from the district service, taking only what it lacks and never overwriting newer data.

3.7 SECURITY AND PRIVACY

Staff sign in with a PIN checked by the relay. Five wrong attempts lock the account for five minutes, and sessions use signed tokens valid for twelve hours. A device holds PIN hashes only for staff of its own facility, and a patient's record reaches only the devices of the registering facility, the facilities on the patient's referrals and district officers. Idle screens lock behind the PIN after fifteen minutes, and on

Android screenshots are blocked. The district service refuses revoked accounts, limits request size and logs every request for identified records, including refused ones.

3.8 DISTRICT OVERSIGHT

The District Health Officer's Command Centre shows referral counts, pending emergencies, escalation alerts, bed occupancy and a map of the facility network. The Audit Trail lists who read or uploaded identified records, and the Data Inspector shows what the district service holds. Every number on these screens comes from a record or says 'not recorded'; no figure is invented.

CHAPTER 4

TEST AND IMPLEMENTATION

4.1 TEST PLAN

Testing was carried out at three levels: unit-level verification suites for each module, browser-based end-to-end checks of complete user journeys, and continuous integration that runs every check on each change pushed to the repository.

4.1.1 SIGN-IN AND PERMISSIONS

Objective: Verify that only valid staff can sign in and that each role sees only its own screens and data.

Procedure: Sign in with correct and wrong PINs; attempt to open restricted screens and server routes with each of the six roles.

Results: Valid PINs open the role dashboard; five wrong PINs lock the account; restricted screens and routes are refused.

4.1.2 TRIAGE ENGINE

Objective: Confirm that triage results are correct and that danger signs are never missed.

Procedure: Run 20 clinical test cases through all three decision paths: model and rules, rules only, and model unavailable.

Results: Every case gives the expected acuity; danger-sign rules raise urgency in every path and never lower it.

4.1.3 REFERRAL FLOW

Objective: Verify that a referral reaches the receiving facility and moves through its life cycle.

Procedure: An ANM registers a RED patient and refers them; the receiving MO opens the Pre-Arrival Board and accepts.

Results: The patient appears on the board of the receiving facility only, and each status change is shown to both sides.

4.1.4 OFFLINE OPERATION

Objective: Confirm that the installed app works without a network and that no data is lost.

Procedure: Install the app, stop the server, open every ANM screen and register patients; restore the network.

Results: All screens open offline; queued uploads are sent oldest first once the network returns.

4.1.5 ACCESSIBILITY TESTING

Objective: Check the pages against WCAG 2.1 level A and AA as required by GIGW for government portals.

Procedure: Run axe-core on 28 pages, signed in with each role.

Results: No WCAG 2.1 A/AA violation is reported on any of the 28 pages.

4.2 TEST CASES

Table 4.1: Test cases and results

S.No	Test case	Expected result	Result
1	Sign in with staff ID and PIN	Role dashboard opens	Pass
2	Enter wrong PIN five times	Account locked for five minutes	Pass
3	Register patient with danger signs	Triage result is RED with reasons	Pass
4	Open a screen without permission	Access refused for that role	Pass
5	Send referral to a higher facility	Patient shown on receiving Pre-Arrival Board	Pass
6	Leave emergency unacknowledged	Escalated to the DHO	Pass
7	Use installed app with server stopped	Screens open; uploads wait in outbox	Pass
8	Deactivate a Medical Officer	Refused by relay and district service	Pass
9	Restart the relay	All accounts and referrals kept	Pass
10	Sign in on a replacement phone	Facility records restored	Pass
11	DHO opens Audit screen	Every record access is listed	Pass

12	Run axe-core on 28 pages	No WCAG 2.1 A/AA violation	Pass

4.3 IMPLEMENTATION

The front end was built as a Next.js application exported as static files, so that it can be served from any web server and cached by the service worker. Application state is held in Zustand stores and persisted to IndexedDB. The triage model is trained in the browser from a seeded synthetic dataset and runs entirely on the device. The mesh relay is a Node.js server using Socket.io; in production mode it keeps staff accounts and referrals in transit in an atomically written store file and refuses to start without its required settings. The district record service is a FastAPI application using SQLAlchemy with SQLite on one machine or PostgreSQL in a district deployment. The Android version is produced with Capacitor.

4.4 IMPLEMENTATION CHALLENGES

4.4.1 UNRELIABLE CONNECTIVITY

Challenge: Uploads fail or stop part-way when the network drops.

Solution: A durable outbox that retries every item, oldest first, and never drops a record.

4.4.2 PATIENT SAFETY IN TRIAGE

Challenge: A model trained on synthetic data could under-triage a critical patient.

Solution: Deterministic danger-sign rules that can only raise urgency, and a slip stating the clinician decides.

4.4.3 PRIVACY OF PATIENT RECORDS

Challenge: Identified records must reach only the facilities that need them.

Solution: Facility-scoped delivery, role and posting checks on every route, and an access log.

4.4.4 LOST OR REPLACED DEVICES

Challenge: A lost phone would take its facility's history with it.

Solution: A replacement phone restores its facility's records from the district service.

4.5 LESSONS LEARNED

- Designing for offline use from the start is far easier than adding it later.
- Clinical safety rules must be deterministic and tested on every change.
- Every number shown to a decision-maker must come from a real record.
- Automated browser tests catch problems that unit tests miss.

4.6 CONCLUSION

Testing and implementation confirmed that NalamMesh works as designed. All planned test cases passed, and the challenges of connectivity, safety, privacy and device loss were addressed with specific mechanisms that are themselves covered by automated checks.

CHAPTER 5

RESULTS AND DISCUSSION

The NalamMesh application was implemented successfully and all modules worked as expected. This chapter presents the main output screens and discusses the results.

5.1 STAFF LOGIN

5.1.1 RESULTS

The Healthcare Staff Portal sign-in page lets a staff member choose the role, the posting and the staff name and enter the PIN. In the evaluation build, demonstration accounts for each of the six roles are listed beside the form, and a banner on every page states that the data is fictional.

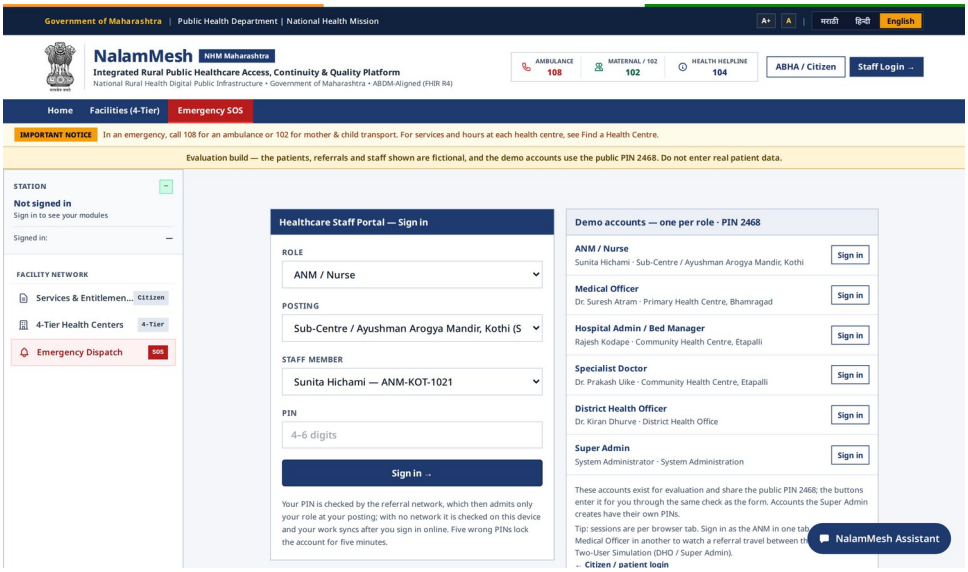


Figure 5.1: Staff Login page

5.1.2 IMPLICATIONS

PIN-based sign-in is quick to use on a shared device, while the relay's lockout and signed tokens protect accounts. Because each device holds only its own facility's PIN hashes, a lost phone exposes far less.

5.2 DISTRICT COMMAND CENTRE

5.2.1 RESULTS

After sign-in, the District Health Officer sees the District Health Command with referral counts, referrals awaiting response, pending emergencies, escalation alerts, beds free and a live map of the facility network with referrals in motion.

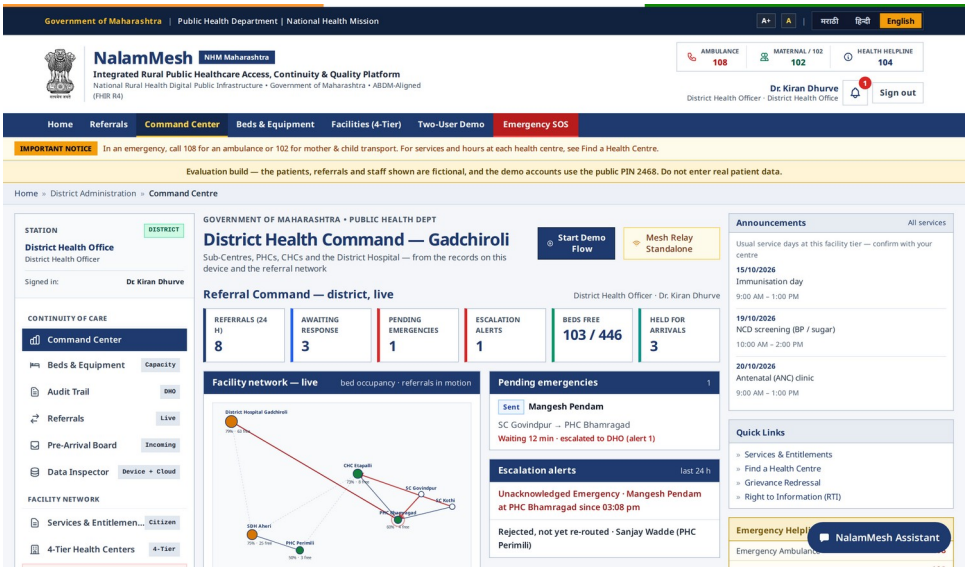


Figure 5.2: District Command Centre

5.2.2 IMPLICATIONS

For the first time, the DHO can see in one place which emergencies are waiting and where beds are available, and act on escalations instead of discovering delays afterwards.

5.3 OPD REGISTRATION AND TRIAGE

5.3.1 RESULTS

The OPD screen collects patient demographics, ABHA ID and vitals in numbered sections. The triage panel shows the RED, YELLOW or GREEN result with reasons, and the live OPD queue board on the same page lists waiting patients.

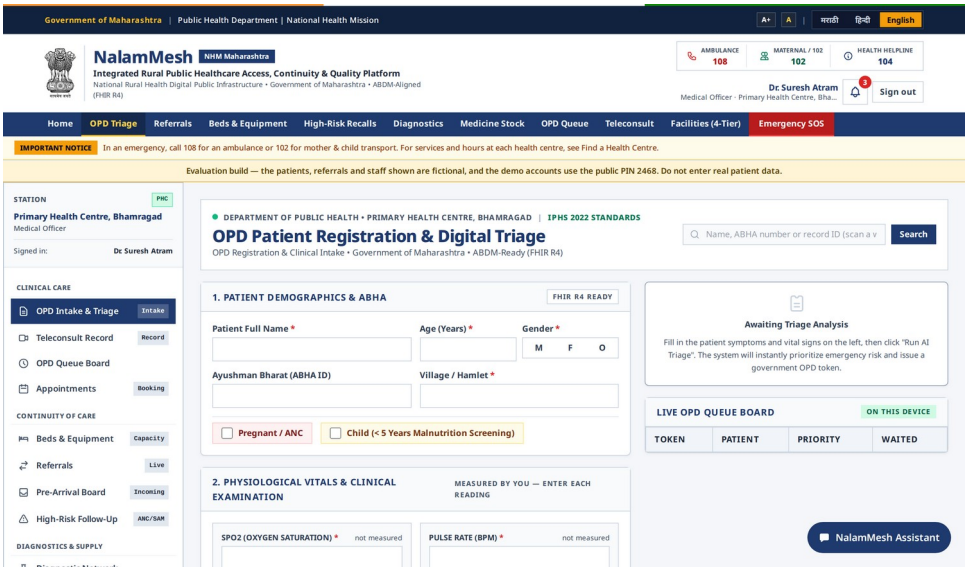


Figure 5.3: OPD Patient Registration and Digital Triage

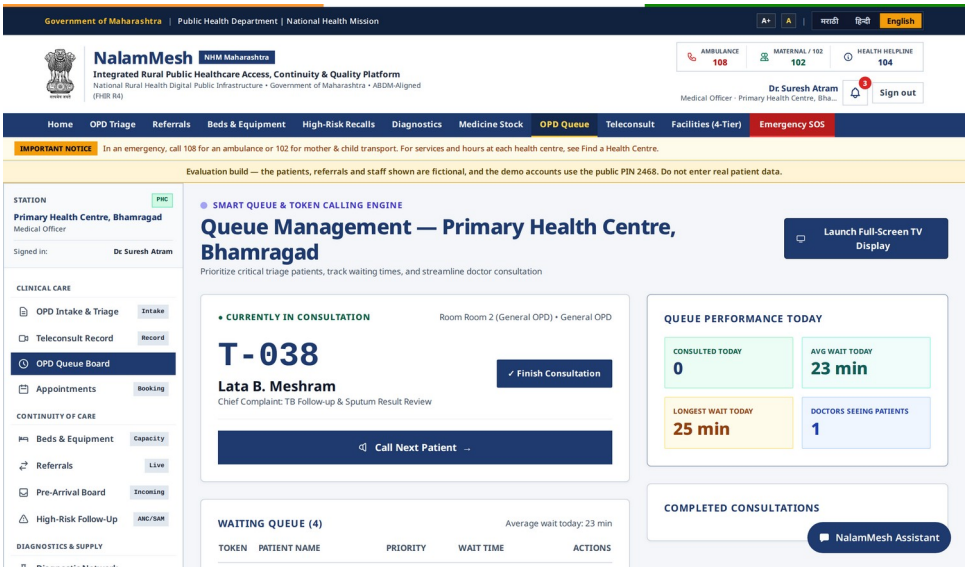


Figure 5.4: OPD Queue Management

5.3.2 IMPLICATIONS

Structured entry ensures that vital signs are recorded for every patient, and consistent triage reduces the chance that a danger sign is missed by a busy worker.

5.4 REFERRALS AND PRE-ARRIVAL BOARD

5.4.1 RESULTS

The Referrals page shows incoming and outgoing referrals in columns for Awaiting response, Accepted, At receiving facility and Closed. The Pre-Arrival Preparation

Board shows patients en route to the facility; when the district service cannot be reached it says so clearly instead of showing an empty list.

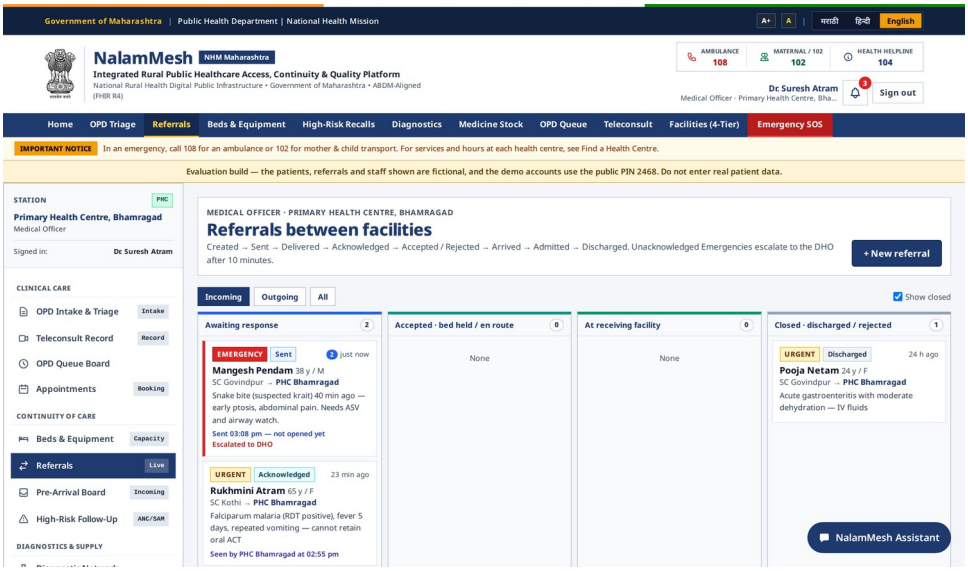


Figure 5.5: Referrals between facilities

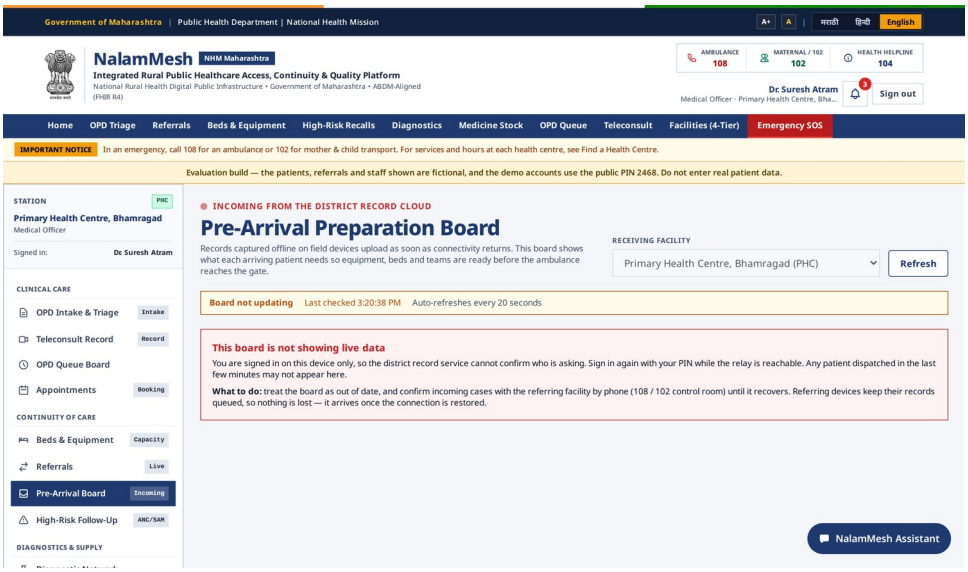


Figure 5.6: Pre-Arrival Preparation Board

5.4.2 IMPLICATIONS

The receiving facility can prepare before the patient arrives, and the honest 'not updating' message ensures staff fall back to a phone call rather than assuming no patient is coming.

5.5 ANM DASHBOARD AND AUDIT TRAIL

5.5.1 RESULTS

The ANM dashboard lists patients registered today, referrals awaiting reply, actions needed and follow-ups due. The Accountability Audit Trail shows every recorded action with who did it, at which facility, on what entity, and the values before and after.

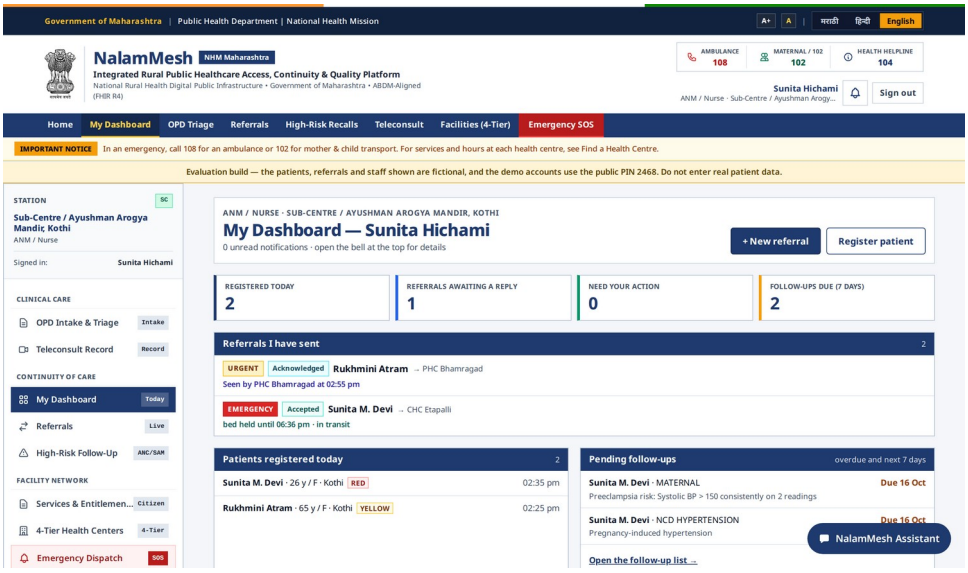


Figure 5.7: ANM Dashboard

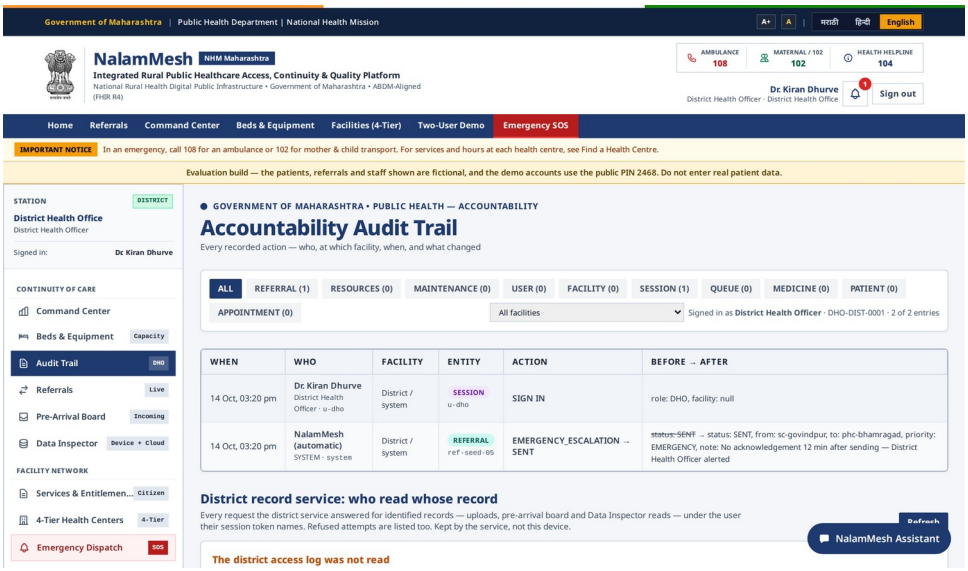


Figure 5.8: Accountability Audit Trail

5.5.2 IMPLICATIONS

Health workers can follow their patients to the end of care, and the department can show who accessed each record, which supports obligations under the Digital Personal Data Protection Act, 2023.

5.6 DISCUSSION

The results show that an offline-first design can provide the benefits of digital records in places where online systems fail. All test cases passed, the full production flow ran successfully in a browser, and the accessibility check reported no violations. Before use with real patients, the triage engine requires clinical validation and regulatory review, and the deployment requires a security audit, government hosting and ABDM certification; these steps are documented in the project's production readiness guide.

CHAPTER 6

CONCLUSION AND FUTURE SCOPE

6.1 CONCLUSION

NalamMesh provides a reliable, offline-first and accountable way to register, triage and refer patients across rural health facilities. By replacing paper registers and referral slips with a single application that works without a network, the system ensures that every patient's record is captured with complete details, travels with the referral and reaches the receiving facility before the patient does. Decision-support triage with safety rules, escalation of ignored emergencies, the Pre-Arrival Board and the access log make the system practical for health workers and trustworthy for the department. The project met all its objectives, and every tested module worked correctly.

6.2 FUTURE SCOPE

NalamMesh is ready for a supervised pilot, and several enhancements can extend its value:

Clinical Validation: A study comparing the tool's triage with clinicians' decisions on real presentations, with ethics approval, followed by review of every threshold by a clinical committee.

ABDM Integration: Production ABHA verification and FHIR record exchange after ABDM certification.

Regional Languages: Translation of all screens into Tamil, Hindi, Marathi and other languages through Bhashini, with clinical terms reviewed by clinicians.

Consent Management: Recording per-patient notice and consent under the Digital Personal Data Protection Act, 2023.

District Analytics: Sending de-identified data to the reporting service for analysis of referral delays and outcomes.

REFERENCES

1. Ministry of Health and Family Welfare, Indian Public Health Standards (IPHS) 2022, Government of India, 2022.
2. National Health Authority, Ayushman Bharat Digital Mission (ABDM) - Health Data Management Policy and ABHA Guidelines, https://abdm.gov.in.
3. World Health Organization, Emergency Triage Assessment and Treatment (ETAT): Manual for Participants, WHO, Geneva, 2005.
4. Government of India, The Digital Personal Data Protection Act, 2023, Ministry of Law and Justice, 2023.
5. Vercel, Next.js Documentation, https://nextjs.org/docs.
6. D. Smilkov et al., TensorFlow.js: Machine Learning for the Web and Beyond, Proceedings of Machine Learning and Systems (SysML), 2019.
7. Socket.IO, Socket.IO Documentation, https://socket.io/docs.
8. S. Ramirez, FastAPI Documentation, https://fastapi.tiangolo.com.
9. Mozilla Developer Network, IndexedDB API and Service Worker API, https://developer.mozilla.org.
10. W3C, Web Content Accessibility Guidelines (WCAG) 2.1, https://www.w3.org/TR/WCAG21.
11. M. Kleppmann, A. Wiggins, P. van Hardenberg and M. McGranaghan, Local-first software: You own your data, in spite of the cloud, Proc. ACM Onward!, 2019.